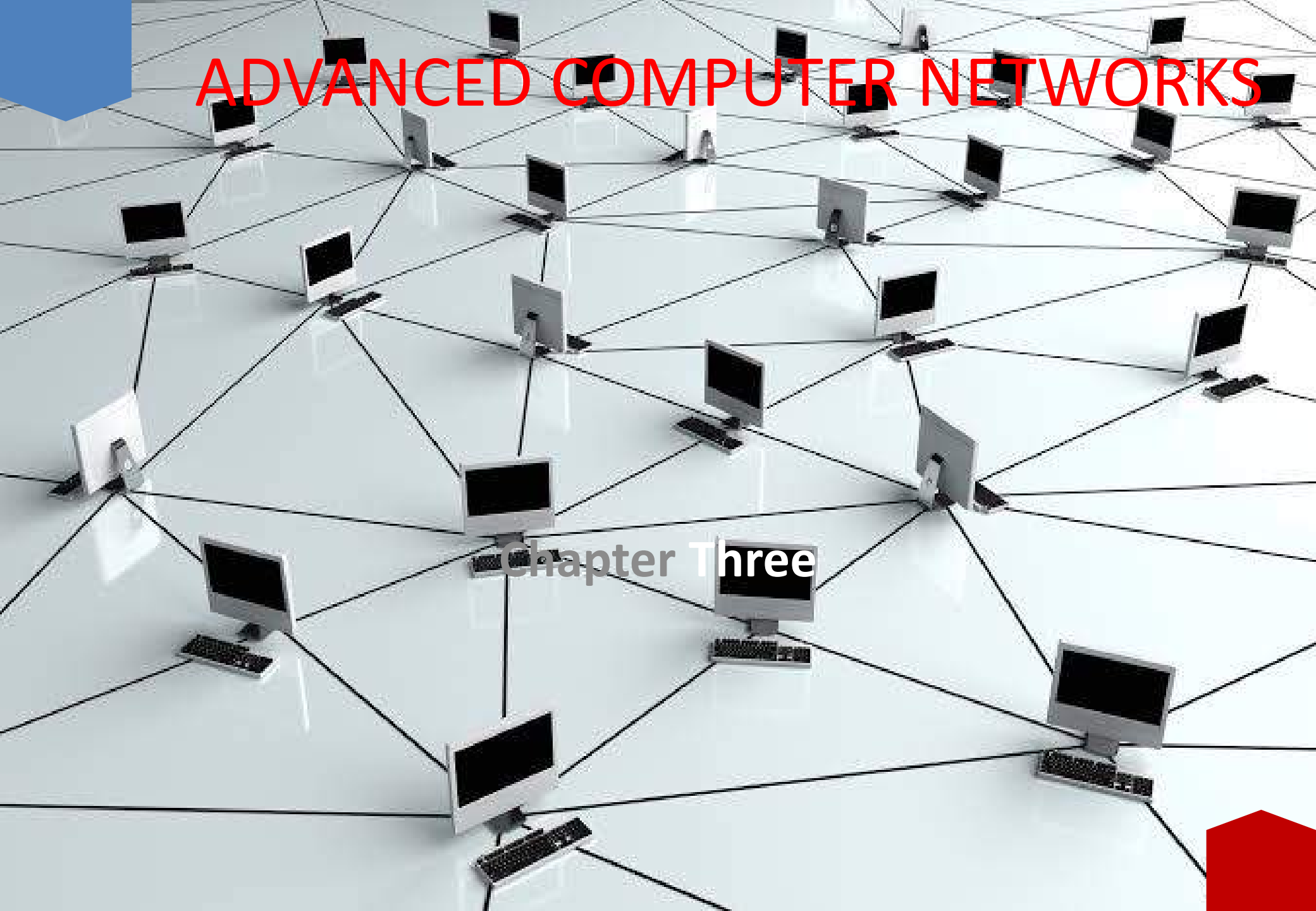
ADVANCED COMPUTER NETWORKS



Chapter Three

INTRODUCTION

An *advanced* computer network is a system that enables computers and other devices to communicate, share data, and access resources through standardized communication protocols and models. Communication models provide guidelines that ensure devices from different manufacturers can communicate effectively

Network Communication model

A **network communication model** is a framework that explains how data is transmitted from one computer (sender) to another computer (receiver) through a network.
It divides communication into different layers, where each layer performs specific tasks.

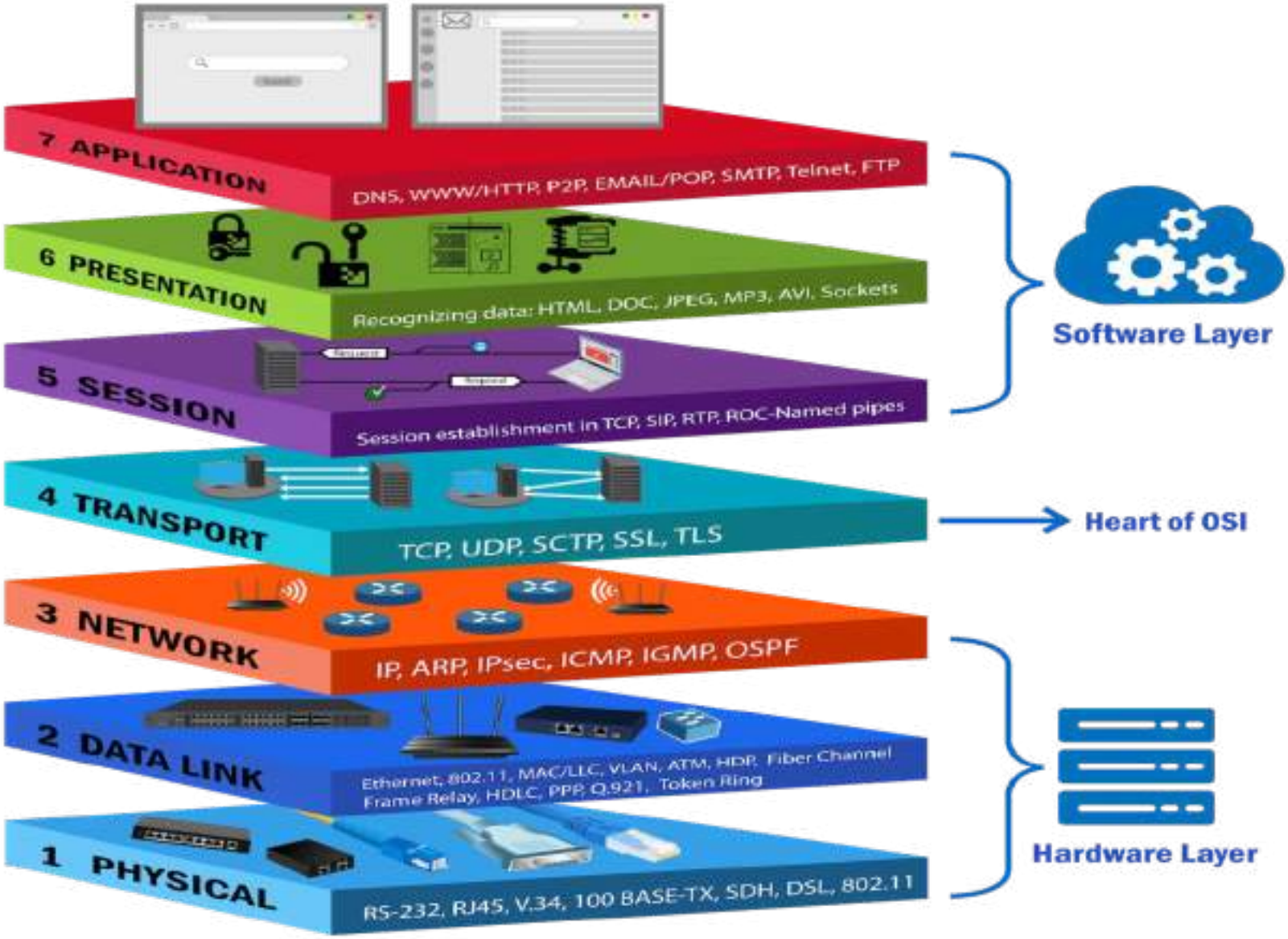
Why Communication Models are Important

- ✓ Standardize network communication.
- ✓ Allow different devices and software to communicate.
  Simplify troubleshooting.
- ✓ Improve compatibility among networking equipment.
- ✓ Help in designing and maintaining networks

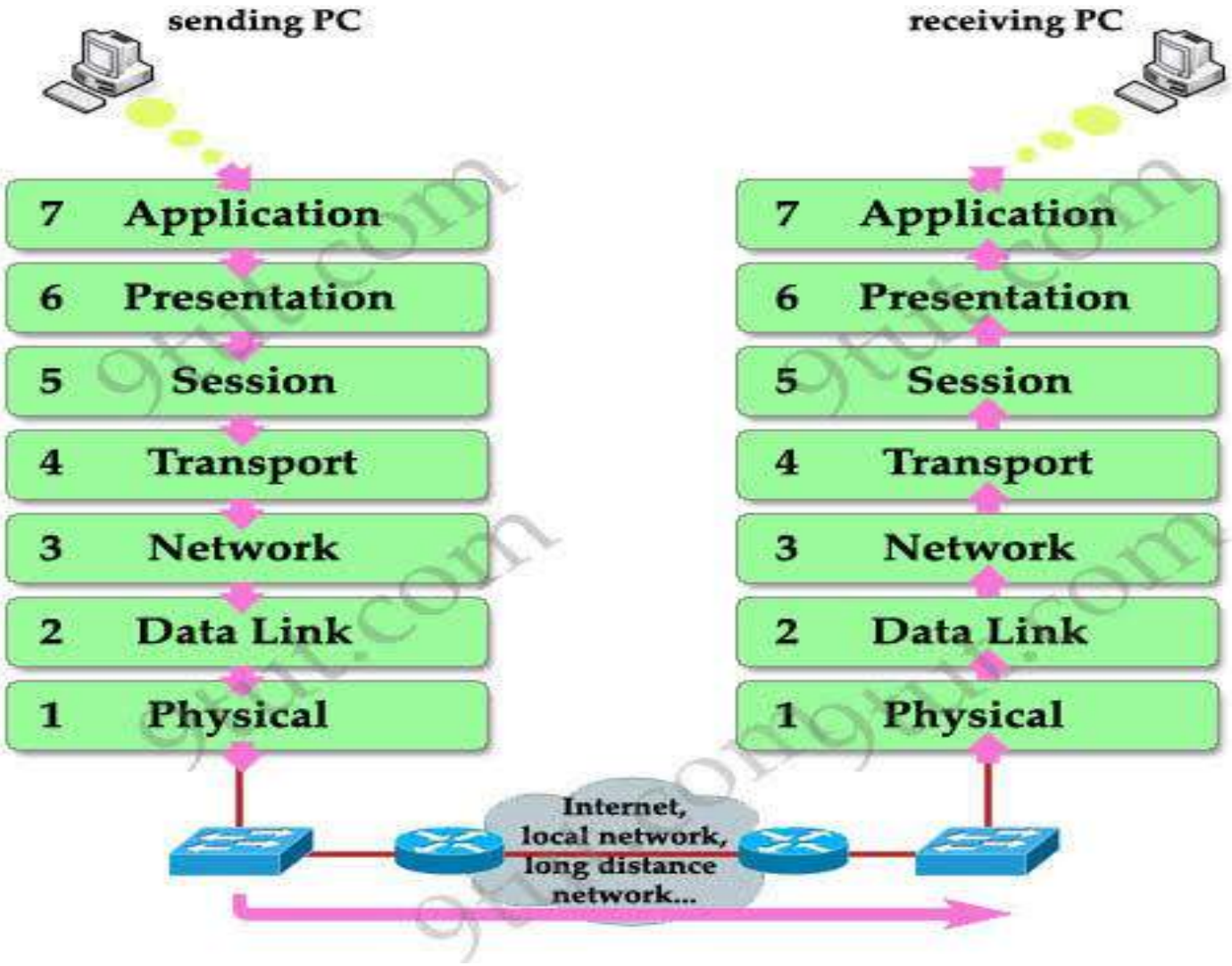
The *OSI* Model

- The **Open Systems Interconnection (OSI) Model** is a conceptual framework developed by the International Organization for Standardization (ISO) for network communication and protocols defination.
- It allow deferent platforms and system to communicate seamlessly. (different OS, Hardware and Software)
- It consists of **seven layers**, each responsible for a specific networking function

The Open Systems Interconnection (OSI) model



Cont..


sending PC
receiving PC
7 Application
6 Presentation
5 Session
4 Transport
3 Network
2 Data Link
1 Physical
7 Application
6 Presentation
5 Session
4 Transport
3 Network
2 Data Link
1 Physical
Internet, local network, long distance network...

ADVANTAGES OF OSI MODEL

flexibility

It is easy to work on each layer(modify them/fix) independently without affect other layer, also easy to adopt the new technologies and protocol.

Conceptual Frame work

The OSI model gives clear (blue print)frame work on how your nodes will communicate

Intra-connectivity/ modular desgn

The OSI model ease the communication of device from different vendors, for they all follow same protocols.

Maintanance/ trouble shooting

In OSI model it is easy to identify faults in the network and fix it by just understanding each layer

~~DISADVANTAGES~~ OF OSI MODEL

No parallel processing

Layers cannot work simultaneously; data must move sequentially down and up through every single layer

Service duplication

The same tasks, such as error and flow control, are repeated across different layers like the data link and transport layers.

Redundant layers

Managing and setting up systems based strictly on all seven layers is slow, cumbersome, and expensive.

Theoretical concept

It is a reference framework rather than a working network architecture, meaning real-world networks rely instead on the practical TCP/IP model.

High complexity

The session and presentation layers offer very little practical utility or distinct functions in most everyday network setups.

Poor timing

It arrived after TCP/IP protocols were already established and widely adopted by the industry.

The communication flow in OSI model

1. Encapsulation (The Sending Process)

When you send data (like an email), it starts at the top layer and moves downward. Each layer wraps the data from the layer above it with its own header (and sometimes a trailer), creating a **Protocol Data Unit (PDU)**

Application, Presentation, & Session Layers: The user interacts with software to generate raw **Data**. These layers format, encrypt, and manage the communication session.

Transport Layer: Breaks the data into smaller pieces and adds a header containing port numbers. This forms a **Segment** (TCP) or **Datagram** (UDP).

Network Layer: Adds a header containing the source and destination IP addresses. This forms a **Packet**.

Data Link Layer: Adds a header with physical hardware addresses (MAC addresses) and a trailer for error checking. This forms a **Frame**.

Physical Layer: Converts the entire frame into electrical, optical, or radio signals. It transmits these **Bits** across the network cables or wireless medium.

2. De-encapsulation at the Receiver

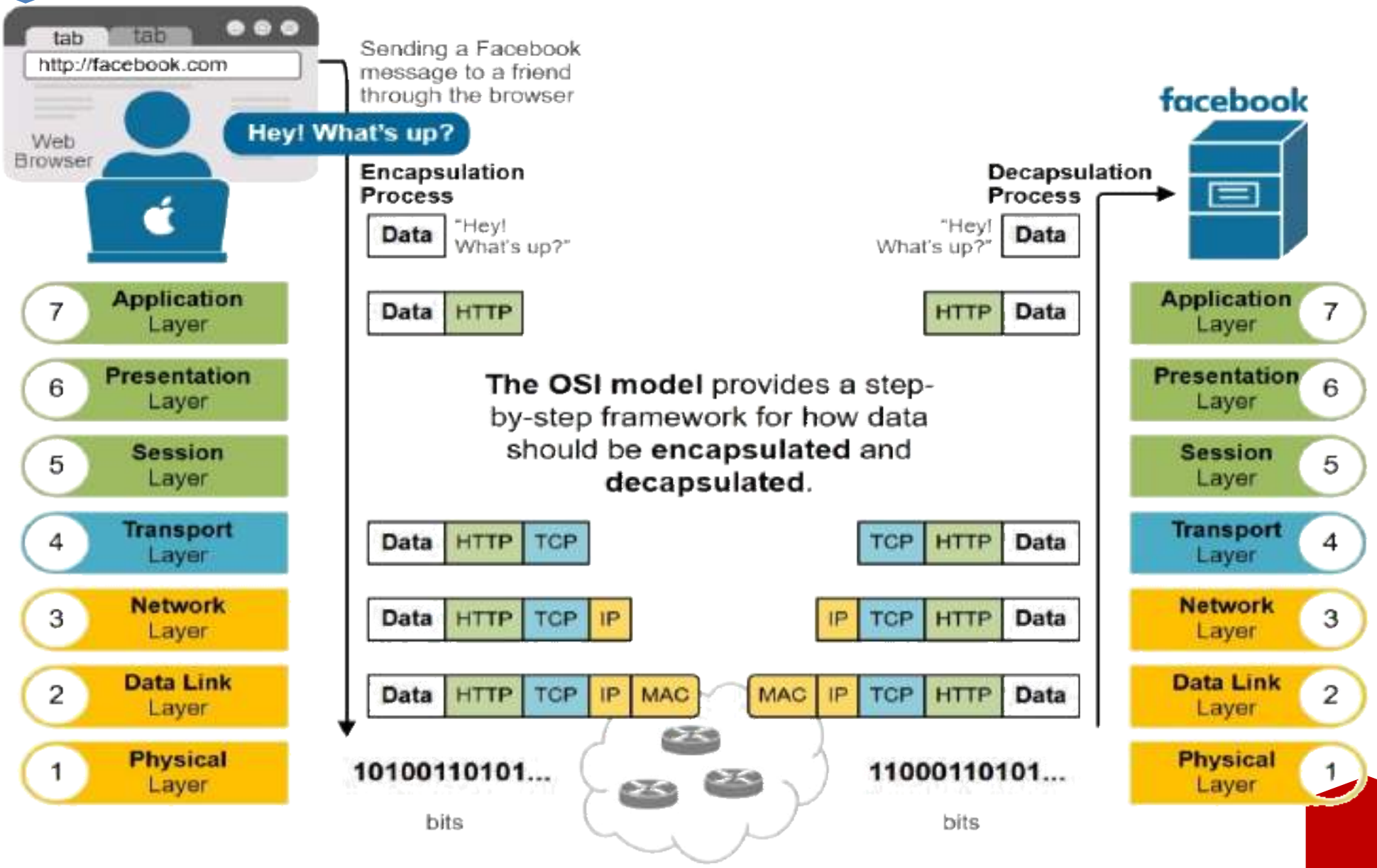
When the other device gets the physical signals, it does the exact opposite process called **de-encapsulation**. It moves up from Layer 1 to Layer 7, stripping off each header and trailer one by one until it recovers the original application data.

To sum up……

As data travels down the sender's computer, each layer adds its own specific set of instructions:

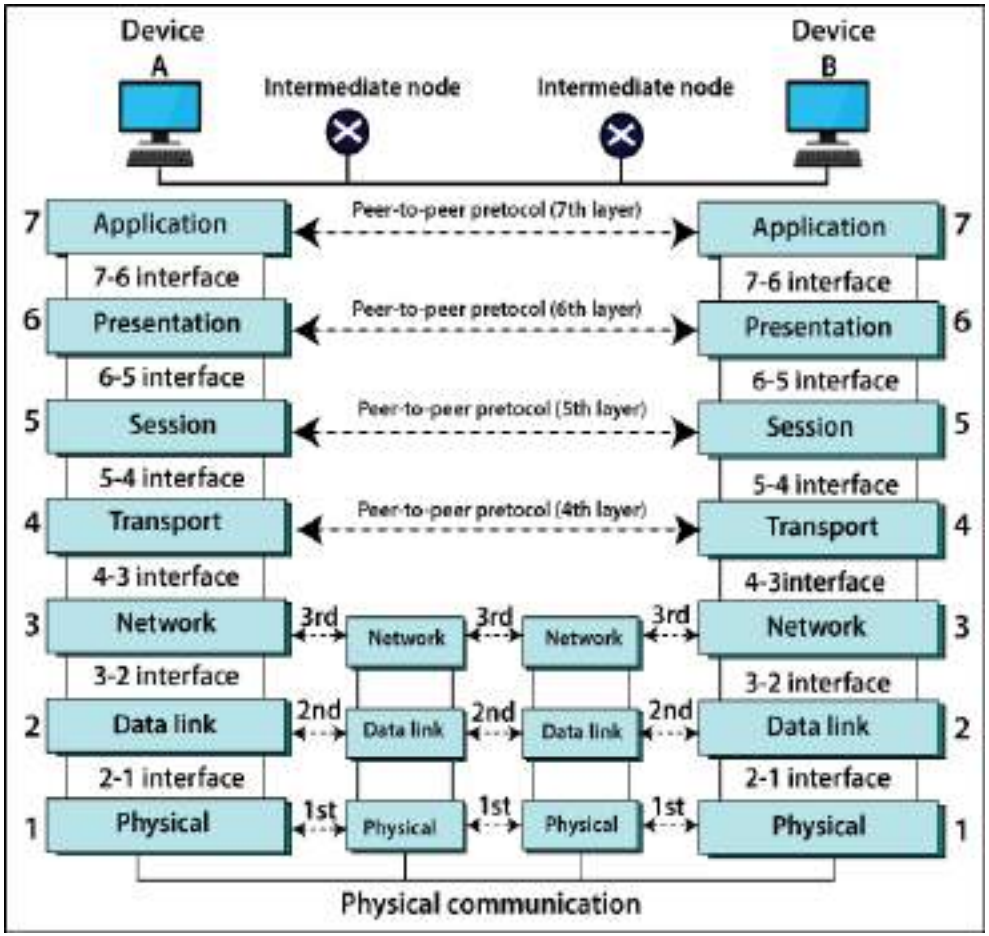
- **Layer 7 (Application):** The user creates raw data or a message (like an email or web request).
- **Layer 6 (Presentation):** Data is formatted, compressed, or encrypted so both devices can read it.
- **Layer 5 (Session):** Sets up, maintains, and ends the communication session between the computers.
- **Layer 4 (Transport):** Breaks the data into smaller pieces and adds a header with port numbers and sequence details, creating a **Segment** (or Datagram).
- **Layer 3 (Network):** Adds source and destination IP addresses to create a **Packet**.
- **Layer 2 (Data Link):** Adds MAC addresses and a trailer for error checking to create a **Frame**.
- **Layer 1 (Physical):** Changes the frame into raw physical **Bits** (electrical, light, or radio signals) to send across the network cable or air

Visual presentation of Data encapsulation

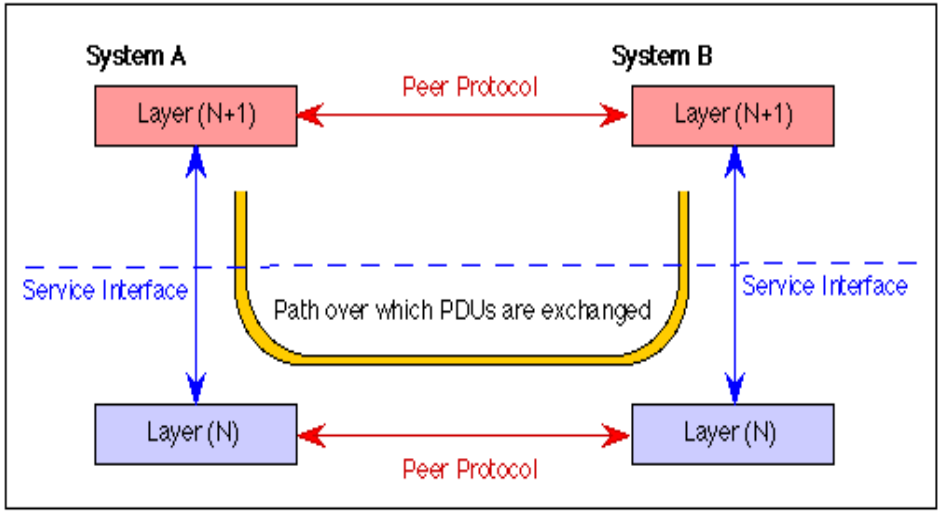


Peer-to-Peer Communication

Peer-to-peer communication in the OSI Model means that a specific layer on a sending device communicates logically with the exact same layer on a receiving device using shared protocols. Key elements include matching layer rules, data packaging, and passing data down and up the stack

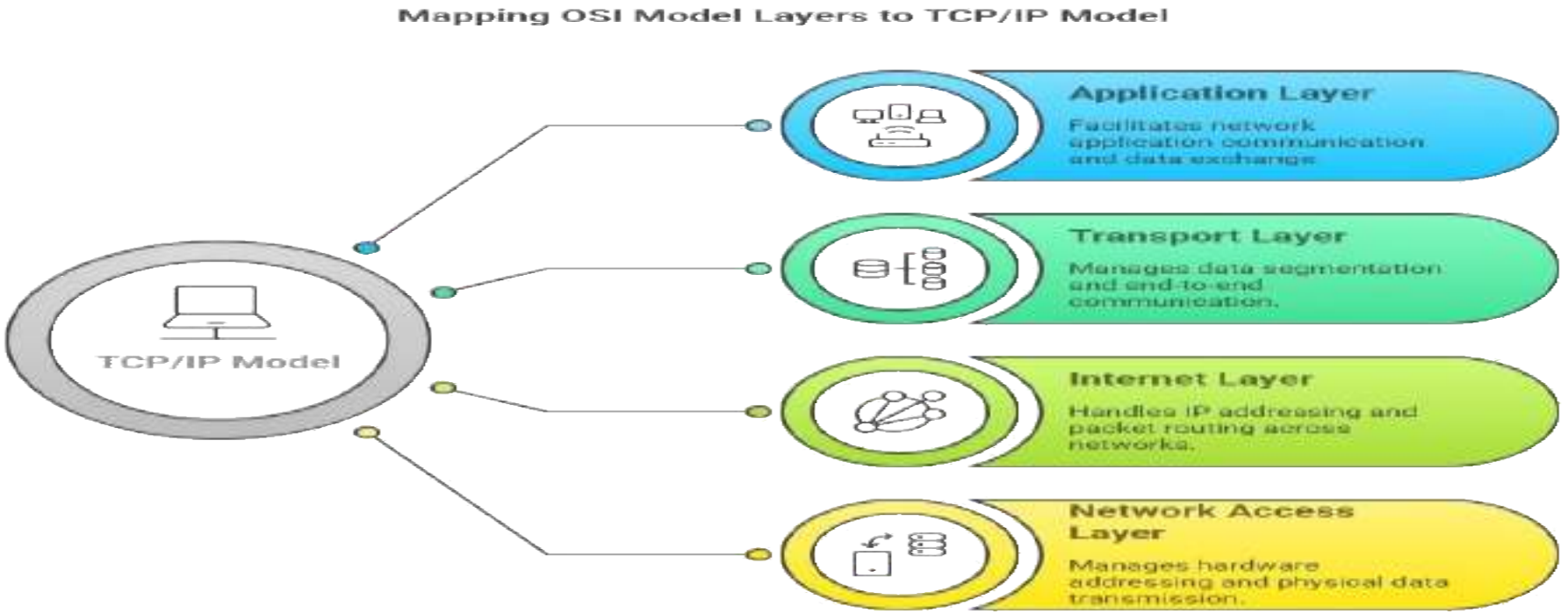


This means layer X communicates with three layers simultaneously, the layer above it, layer below it and the layer at the same level



TCP/IP MODEL

- The TCP/IP (Transmission Control Protocol/Internet Protocol) model is the fundamental communication framework for the internet and most local networks. It organizes data transmission into a standardized, four-layer architecture



Application Layer

Interfaces directly with user software (e.g., web browsers, email clients) using protocols like HTTP and FTP.

Transport Layer

Manages end-to-end data delivery. It uses TCP for reliable, error-checked transmission or UDP for faster, connectionless delivery.

Internet Layer

Uses IP addresses to logically route and forward packets across different networks to their final destination.

Network Access (Link) Layer

Handles the physical hardware and electrical signals required to transmit data over local cables or wireless connections.

Developed by the U.S. Department of Defense, it is the practical, protocol-driven alternative to the theoretical seven-layer OSI model.

OSI Model vs TCP/IP Model



OSI Model		TCP/IP Model
7 Applicaton Network process to application	DHCP, DNS, FTP, HTTP, HTTPS, POP, SMTP, SSH, HTML, JPEG, etc.	Applicaton 4 Consists of applications and processes that use the network
6 Presentation Data representation and encryption		
5 Session Interhost Communication		
4 Transport End-to-end connections and reliability	TCP UDP	Transport 3 Provides end-to-end data delivery services
3 Network Path determination and logical addressing	IP Address : IPv4, IPv6	Internet 2 Defines the datagram and handles the routing of data
2 Data Link Physical addressing	MAC Address	Network Access 1 Network access layer consists of routines for accessing phycial networks
1 Physical Media, signal and binary transmission	Ethernet cable, fibre, wireless, RS-232, RJ45, SDH, DSL, etc...	

Advantages of TCP/IP MODEL 😊

✔**Platform independence**: Works across different types of computer hardware and operating systems.

✔**Scalability**: Easily supports small local networks as well as massive global networks.

✔**Reliable communication**: Uses error checking and tracking to make sure data arrives safely.

✔**Open standard**: Publicly available and not controlled by a single company.

✔**Flexible routing**: Supports multiple paths to move data efficiently

Disadvantages of TCP/IP MODEL

👎 **Poor separation:** It does not clearly separate services, interfaces, and protocols, making it hard to use for new technologies.

👎 **Not generic:** It only fits the TCP/IP suite and cannot easily describe other protocol types like Bluetooth.

👎 **Layer mixing:** It combines data link and physical functions into one layer and lacks the deep detail found in the OSI Model.

👎 **Rigid protocols:** Replacing an old or weak protocol inside the model is very hard to do

👎 **Setup difficulty:** Configuring and managing large TCP/IP networks requires high technical skill.

👎 **Weak security:** The early design lacked built-in security, making it open to attacks like SYN flooding.

👎 **High overhead:** Extra error checks and delivery tags slow down data speed and use more network power.

👎 **No delivery guarantee:** The transport layer uses some parts (like UDP) that do not promise data will arrive safely

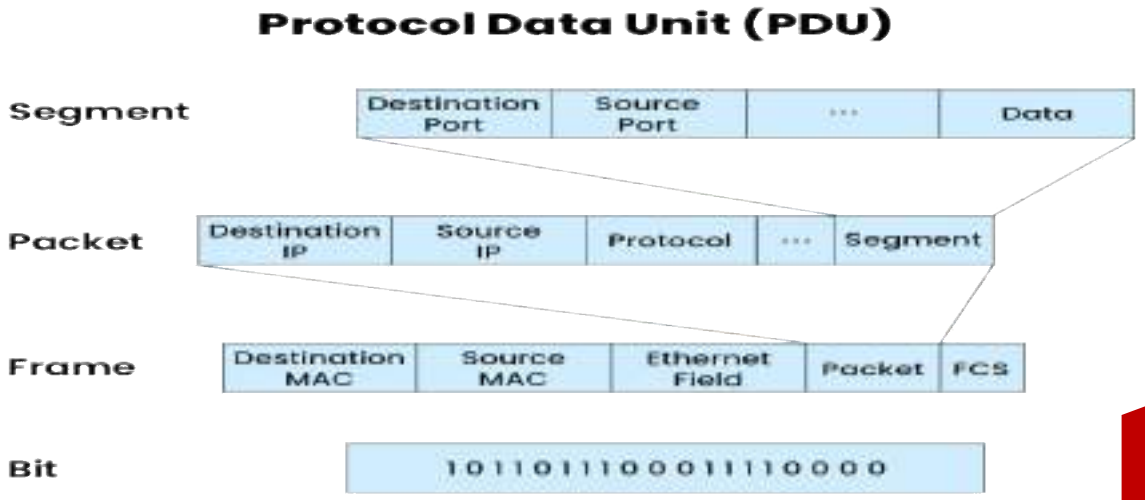
ClassWOrk, Pg130, ex3.1

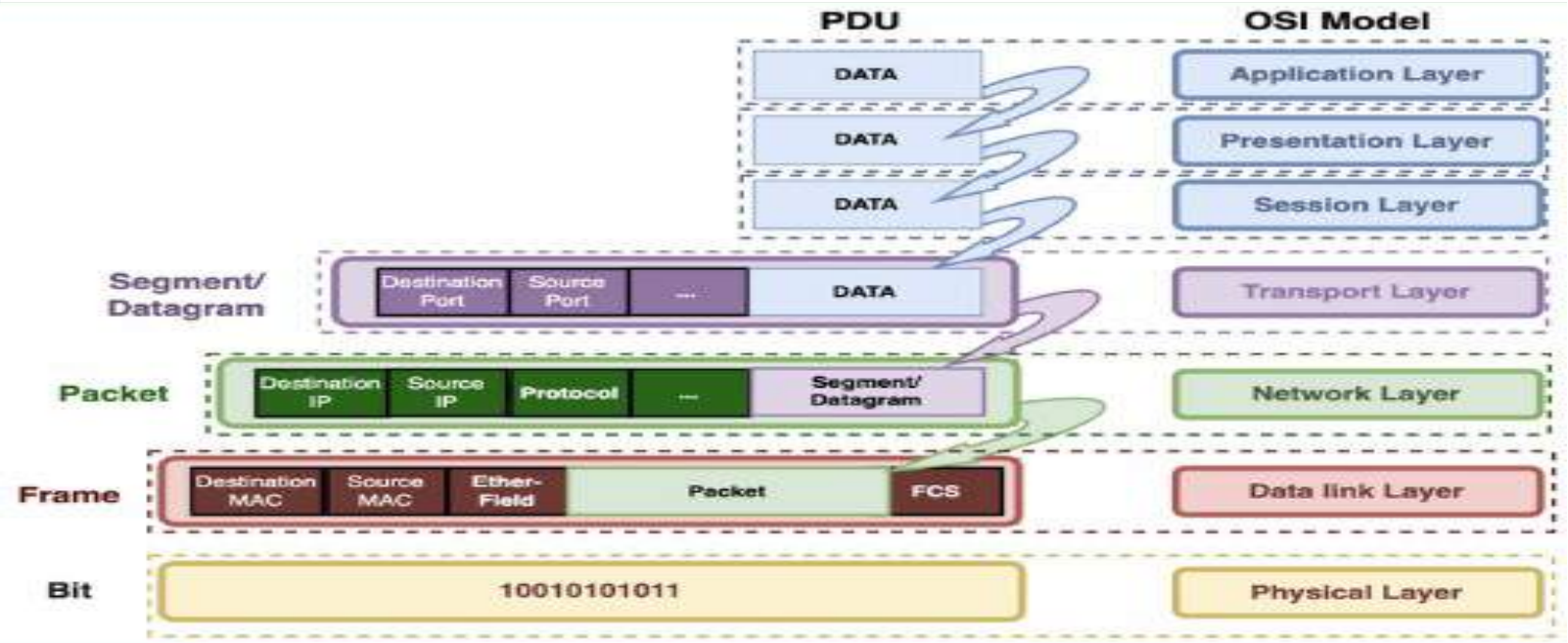
Data **Encapsulation** in Network communication

- Encapsulation in networking is the process of wrapping data with protocol headers and trailers at each OSI model layer, enabling proper routing, error detection, and delivery across LANs, WANs, and the internet; for example, a web request from a browser becomes a series of layered packets, frames, and bits.

To facilitate communication each layer uses *PDU* which controls information attached to the data

a **Protocol Data Unit (PDU)** is the basic unit of data passed between layers of a network model, changing names by layer into **segments**, **packets**, **frames**, or **bits**.





Network addressing and **routing** are the core systems that guide data packets across interconnected computer networks. Addressing assigns logical labels to devices, routing selects the best path for data flow, and routers forward packets between different network segments.

Addressing Concepts

- **IP Addresses**: Logical labels (like IPv4 or IPv6) used to identify source and destination devices across the global internet.
- **Subnet Masks**: Numbers that split an IP address into a network part and a host part to see if a device is local.

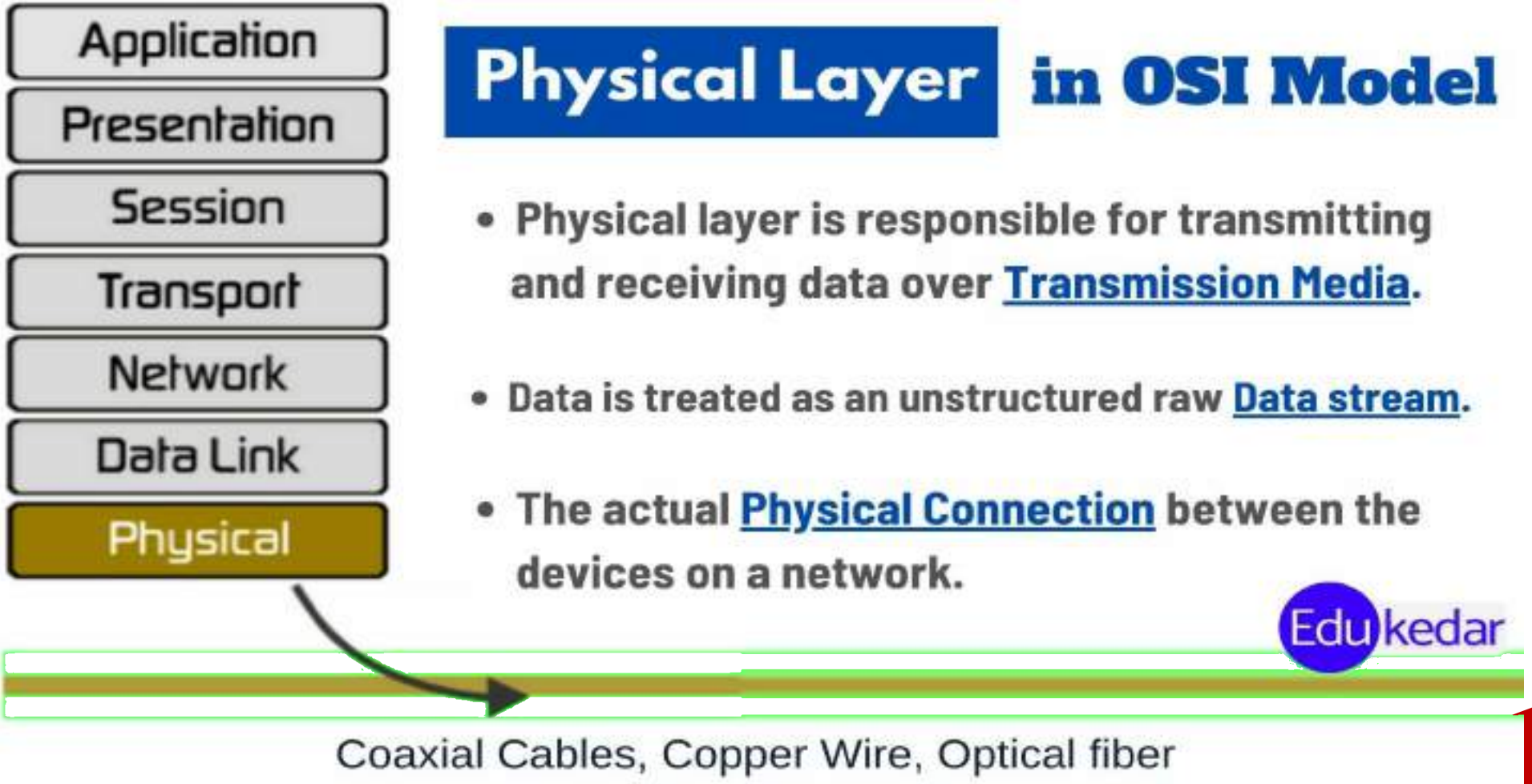
Routing Concepts

- **Routers**: Gateways that connect separate networks and move data packets from origin to destination.
- **Routing Tables**: Maps stored in a router's memory that list destination networks, next hops, and exit interfaces.
- **Static vs. Dynamic**: Routes are either set up by hand (static) or updated automatically by routing protocols like OSPF and BGP (dynamic).

Group work,pg 134,Ex 3.2

LAYER 1:PHYSICAL LAYER

The physical layer is the lowest layer (Layer 1) of the OSI model, responsible for transmitting raw binary data bits across a physical communication medium. It manages hardware components, signal encoding, and data transmission modes.



Core Functions

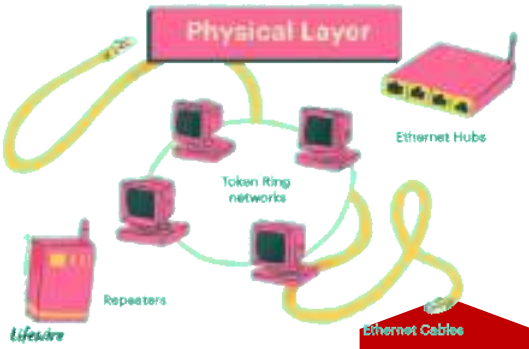
- **Bit Transmission:** Converts data bits (0s and 1s) into electrical, optical, or radio frequency signals.
- **Line Coding & Modulation:** Maps digital bits onto carrier signals using techniques like amplitude or frequency shifting.
- **Bit Synchronization:** Ensures the sender and receiver clocks match for accurate bit reading.
- **Transmission Modes:** Controls data flow direction via simplex, half-duplex, or full-duplexing.
- **Topology & Interfaces:** Defines physical layouts (star, bus, ring) and pin configurations for connectors.

Hardware and Technologies

- **Cabling & Media:** Copper wires (twisted pair)()electrical , fiber optic cables(light), and wireless(radio) airwaves.
- **Network Devices:** Hubs, repeaters, network interface cards (NICs), and modems.
- **Protocols/Standards:** Ethernet (IEEE 802.3), Wi-Fi (IEEE 802.11), and Bluetooth (IEEE 802.15.1)

Characteristics of Physical Layer

- Electrical
- Functional
- Procedual
- Mechanical



Exercise, pg 136,Ex 3.3

LAYER 2: DATALINK LAYER

- Is Layer 2 in the OSI network model, responsible for **node**-to-**node** local delivery, framing, and error control. It sits directly between the physical layer and the network layer.

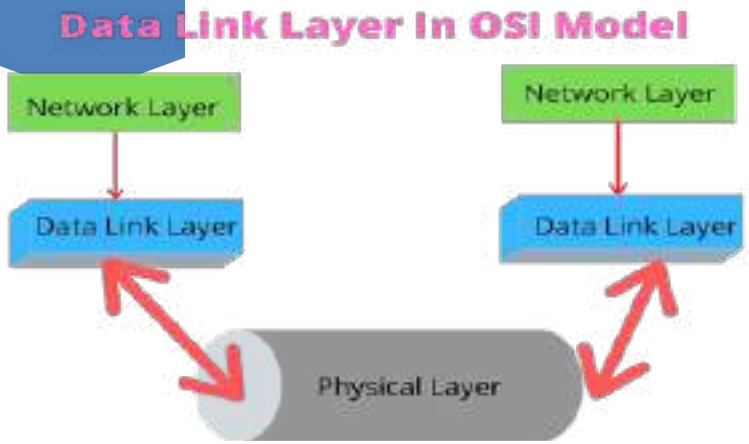
Functions

- **Framing:** Breaks packets from the network layer into smaller pieces called frames.
- **Physical Addressing:** Adds source and destination MAC addresses to frame headers to ensure correct local delivery.
- **Error Control:** Detects and corrects transmission errors using checksums or trailers. (ACK/NACK)
- **Flow Control:** Prevents a fast sender from overwhelming a slow receiver with too much data.
- **Access Control:** Manages how devices share the same physical communication line.

Sublayers

- **Logical Link Control (LLC):** Handles error messages, flow control, and software multiplexing.
- **Media Access Control (MAC):** Manages physical hardware addressing and medium access.

ACK (Acknowledgement) and **NACK** (Negative Acknowledgement) are signal messages sent by a receiving device back to a sending device to report whether data arrived safely or broken



Logical Link Control (LLC)

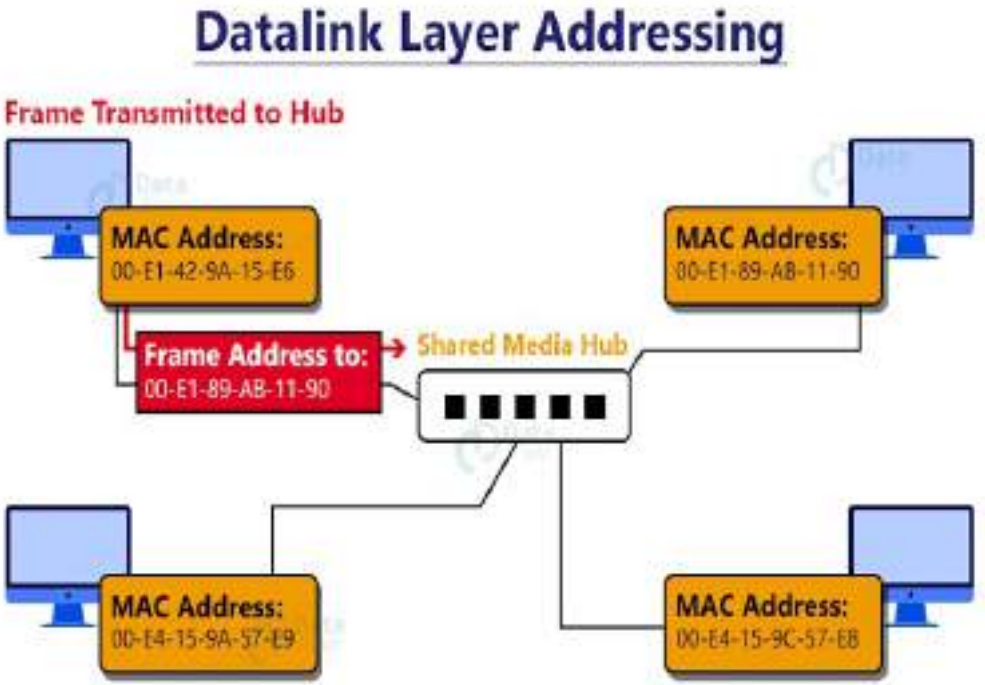
The **LLC sublayer** acts as the interface between the network layer above it and the physical hardware below it

- **Software Interface:** Communicates directly with network layer protocols like IPv4 or IPv6.
- **Multiplexing:** allows multiple network protocols to share the same physical network link simultaneously.
- **Standard:** Defined globally by the IEEE 802.2 specification

Media Access Control (MAC)

The **MAC sublayer** interfaces directly with the physical layer to manage how data is physically sent over the wire or air.

- **Hardware Interface:** Coordinates directly with the physical network interface card (NIC).
- **Physical Addressing:** Adds the unique 48-bit **MAC addresses** of the sender and receiver to each frame.
- **Media Access:** Determines when a device is allowed to transmit data over a shared medium.
- **Collision Avoidance:** Uses protocols like CSMA/CD (Ethernet) or CSMA/CA (Wi-Fi) to prevent data collisions.
- **Framing:** Wraps data with a preamble and a Frame Check Sequence (FCS) for error detection



Switch LAN concept

- A switched Local Area Network (LAN) uses a Layer 2 data link device an Ethernet switch to connect end devices, forward data frames based on MAC addresses, and eliminate shared medium collisions. Key elements include: dedicated switch ports, MAC address tables, and full-duplex communication

a) Link layer addressing and RP

Link Layer Addressing (MAC Addresses)

- **Physical identification:** Uses a unique 48-bit (6-byte) hardware address hardcoded into a device's Network Interface Card (NIC).
- **Local scope:** Destination and source MAC addresses change at every router hop, valid only on the current local network segment (LAN).
- **Frame delivery:** Encapsulates data into frames so local physical hardware can accept or reject the packet

Address Resolution Protocol (ARP)

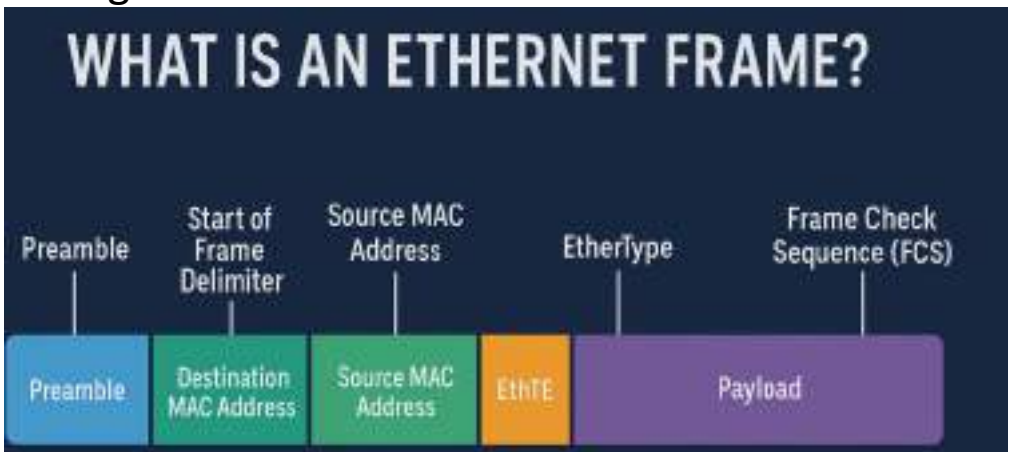
- **Address mapping:** Translates a Layer 3 logical IP address (IPv4) into a Layer 2 physical MAC address.
- **ARP request:** Broadcasts a local message asking *"Who has this IP address? Tell me your MAC address"*.
- **ARP reply:** Responds via unicast with the specific hardware MAC address matching that IP.
- **ARP cache:** Temporarily saves these IP-to-MAC pairings to speed up future local device communication.

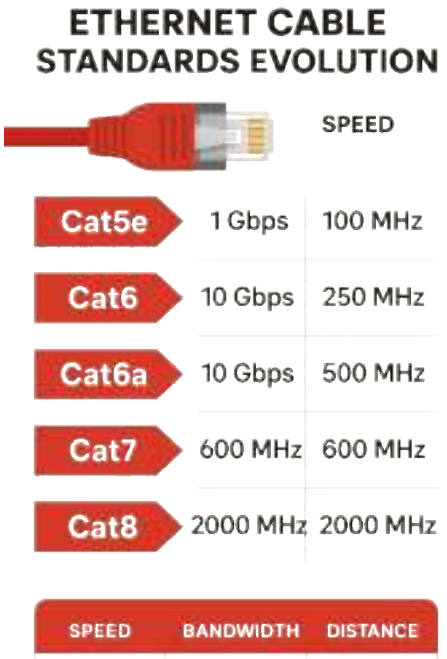
b) **Ethernet**

Ethernet is a foundational networking technology operating across the physical and data link layers of the OSI model, standardizing how wired local area networks transmit data frames like payloads and addresses.

Frame Components

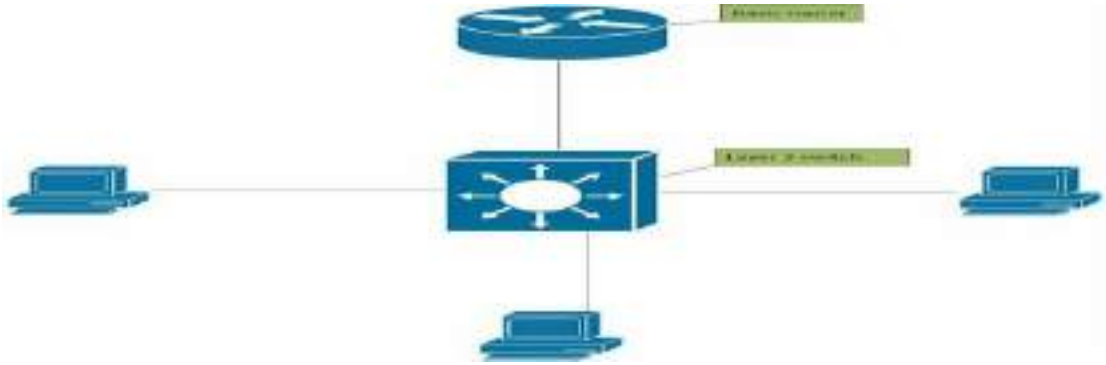
- **Preamble:** A pattern of bits that helps synchronize the receiver's clock.
- **MAC Addresses:** Physical destination and source addresses identifying the receiving and sending hardware.
- **EtherType Field:** Identifies which upper-layer protocol (such as IPv4 or IPv6) is carried in the frame.
- **Payload:** The actual data being transported across the network.
- **Frame Check Sequence (FCS):** An error-detecting code used to ensure the frame was not corrupted during transit.





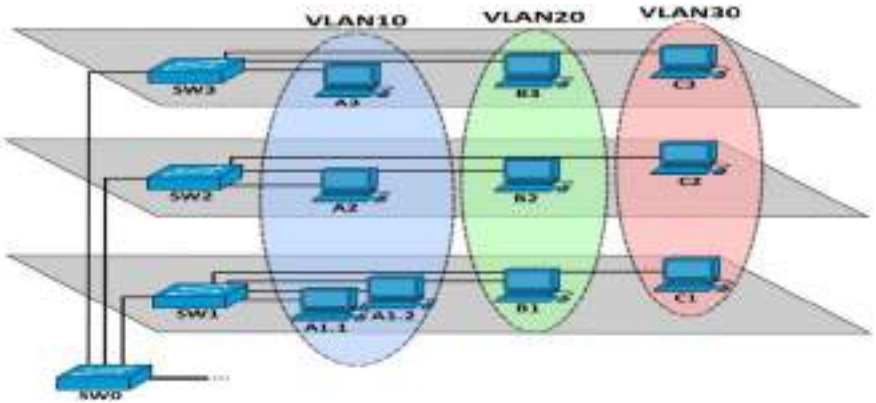
c) link-layer switches

A link layer switch is a Layer 2 networking device that receives data frames and forwards them to specific destination ports using Media Access Control (MAC) addresses. It include self-learning tables, dedicated collision domains, and transparent operation.



d)VLANs (Virtual Local Area Networks)

is a smart way to split one physical network switch into multiple independent virtual networks. Key features include **logical separation**, **better security**, and **reduced broadcast traffic**



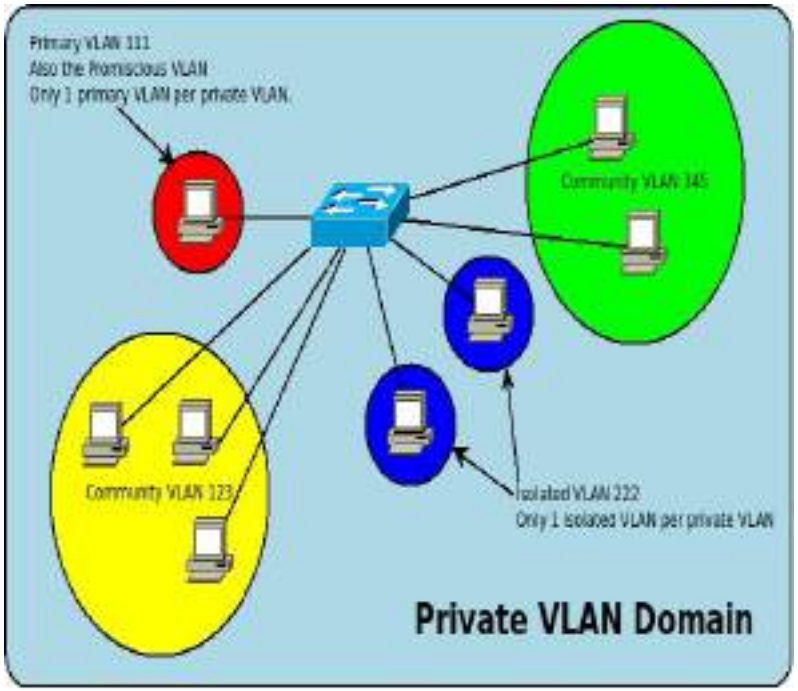
How VLANs Work/configure

- **Logical Split:** Each port on a switch is given specific VLAN to receive frames separately from the internet. Groups devices by software settings instead of physical wires.
- **Broadcast Domains:** Keeps noisy background data (broadcasts) trapped inside its own group.
- **Frame Tagging (IEEE 802.1Q):** Adds a small electronic tag to data packets so switches know which group they belong to.
- **Inter-VLAN Routing:** Requires a router or a Layer 3 switch for different groups to talk to each other

Benefits of VLANS



- **Better Security:** Separates sensitive groups (like HR or Finance) so unwanted users cannot see the data.
- **Smaller Broadcast Domains:** Cuts down on random background data noise to keep the network fast.
- **Traffic Control:** Stops data from going where it does not need to go.
- **Easy Management:** Groups users by job or need instead of where their desks sit in a building.
- **Flexibility:** Moves devices to a different virtual group through software settings instead of moving wires.
- **Lower Costs:** Uses the same physical boxes and cables you already own instead of buying new hardware.

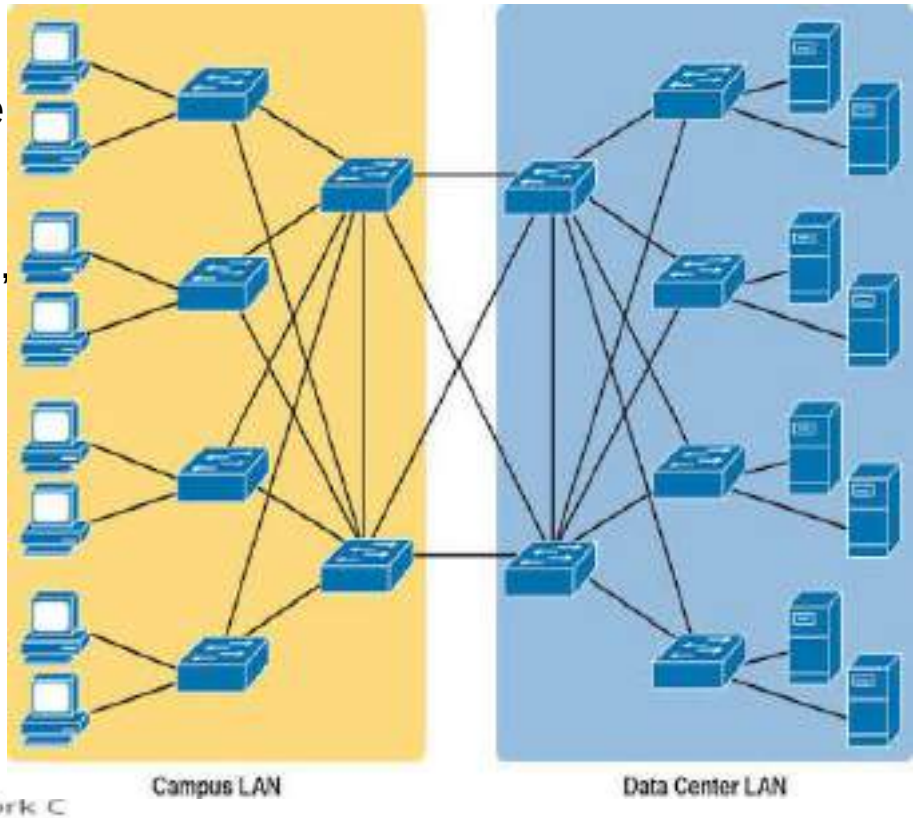


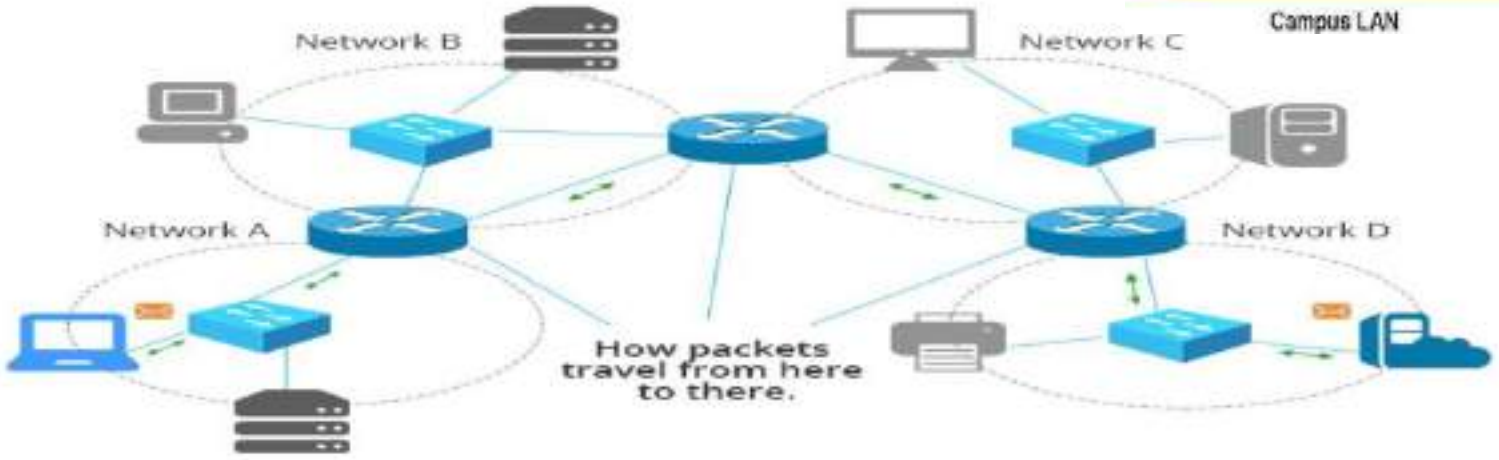
Bottlenect of VLANS

Configuration Complexity: Setting up multiple VLANs requires advanced skills to manage complicated switch configurations.
Troubleshooting Difficulty: Finding and fixing a network issue takes much longer when tracing invisible, logical paths.
VLAN Sprawl: Creating too many distinct networks makes the overall infrastructure messy and hard to manage.
Routing Bottlenecks: Moving data between different VLANs requires a router or Layer 3 switch, which can slow down speeds.
Additional Equipment Costs: High-performance routing hardware is often needed to handle inter-VLAN traffic without lag.
VLAN Hopping Attacks: Malicious users can exploit configuration flaws to jump from a secure VLAN to a private one.
Single Point of Failure: A hardware failure on the main physical switch crashes every single virtual network hosted on it

Key features/x’tics of SWITCHED LANS

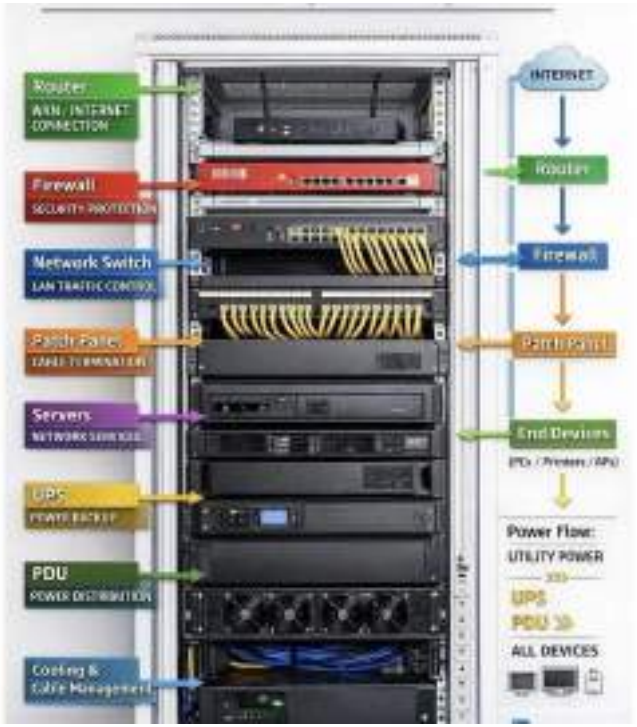
- **MAC Address Learning:** The switch records physical source addresses from incoming frames into an internal table to map devices to specific ports.
- **Direct Forwarding:** It sends data frames straight to the destination port instead of broadcasting them to every device, which stops network congestion.
- **No Collisions/switching mechanism:** Microsegmentation gives every connected device its own dedicated slice of the network, eliminating data collisions entirely.
- **Full-Duplex Data:** Devices can send and receive data at the same time, which doubles the speed and capacity of the connection





Components of SWITCHED LAN

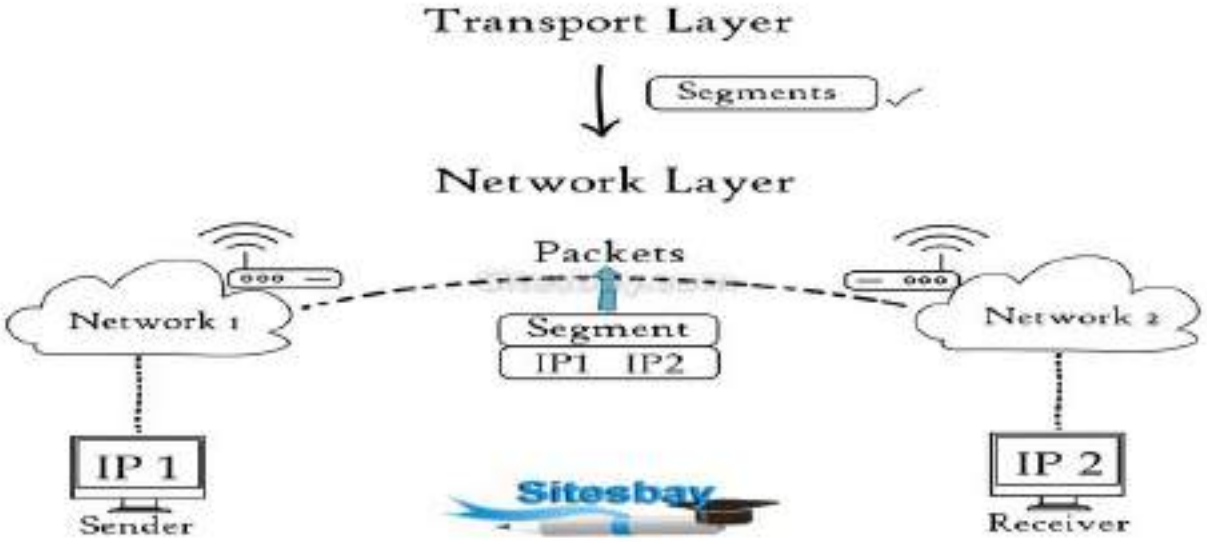
- **Network Switches**:
- Act as the central connection point, using MAC address tables to filter and forward data frames directly to their intended targets.
- **Host Devices**:
- Endpoints like computers, IP phones, and printers that generate or receive network traffic.
- **Network Interface Cards (NICs)**:
- Hardware inside each host that translates data into electrical, optical, or radio signals.
- **Routers**:
- Connect the local switched LAN to other networks or the internet.
- **Access Points (APs)**: Add wireless devices to the wired switched infrastructure.
- **Network Cables**:
- Copper twisted-pair (Ethernet cables like Cat5e/Cat6) or fiber-optic cables used to link devices to switches.
- **Virtual LANs (VLANs)**:
- Software configurations that split a physical switch into separate logical networks for security and traffic control.
- **MAC Address Tables**:
- Memory inside the switch tracking which port connects to which device ID.



Activity and Ex.pg 141,ex 3.4

The Network Layer /Layer 3

- Is **Layer 3** of the OSI model, responsible for logical addressing, routing, and packet forwarding across different networks. Its main components include IP addresses, packets, and routers





key Functions of Network layer

- **Logical Addressing**: Assigns unique IP addresses (like IPv4 or IPv6 ICMP, OSPF, and BGP) to identify source and destination devices.
- **Routing**: Finds the best and fastest path for data to travel across multiple connected networks.
- **Encapsulation**: Wraps data segments from the transport layer into packets by adding source and destination IP headers.
- **Fragmentation**: Breaks large packets into smaller pieces if the receiving network requires a smaller size

NOTE: on P2P connection network layer it is not necessary

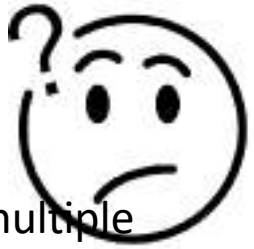
Key takeaway

The bottom three layers of the OSI model foccusses on communication and Connection between networks

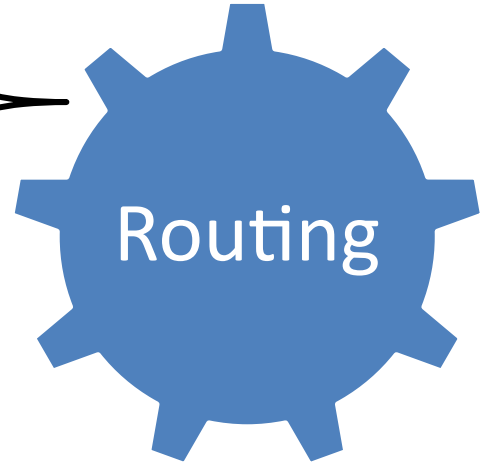
Activity: PG143,Act_3.4

Design a simple coffee shop LAN using a Cisco packet tracer

Network Layer Design Issue



Network layer design focuses on moving packets from a source to a destination across multiple routers. Key challenges include store-and-forward packet switching, defining transport layer services, choosing between connectionless or connection-oriented models, and resolving complex routing and congestion problems.



routers fail to find, maintain, or use the best path to move data packets from a source to a destination.

- **Link State (like OSPF)**: Uses more memory and CPU power to calculate shortest paths across the whole topology map. (OpenShoertestPathFirst)
- **Path Vector (like BGP)**: Misconfigurations can cause global internet traffic blackholes or route leaks(BroadGetawayProtocol)

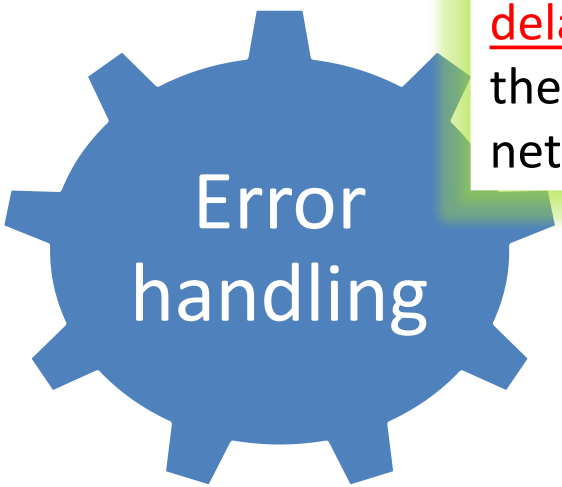


involve errors in IP assignment(Unique IP ID), subnet masking, or packet modification. These problems prevent routers from matching packets to the correct routes, leading to dropped traffic or unreachable hosts



Large data ussually are breaked into smaller packes and devided/routed then assembled again on destination

occur when data packets are delayed, dropped, or corrupted as they *travel independently* across a network



occur when routers fail to detect, report, or respond properly to transmission failures.



deals with managing traffic when the demand for network resources (like router buffer space or link bandwidth) exceeds available capacity.

*congestion control (like TCP sliding windows), network layer control relies on routers to directly monitor and manage traffic loads.

Network layer security issues involve vulnerabilities where attackers manipulate IP packets, routing protocols, or infrastructure to intercept data, disrupt service, or bypass access controls.



Harden Firewalls and ACLs AccessControlList

Deploy IPsec (IP Security):

Authenticate Routing Protocols(cryptographibg)



IPv4 to IPv6 Incompatibility

occur when different networking technologies, protocols, or architectural standards fail to communicate or exchange packets seamlessly.